SE-2

Space Edge Two

SPIRAL BLUE

Based on the NVIDIA Jetson series, SE-2 is the latest onboard computing system developed by Spiral Blue that enables on board edge processing and artificial intelligence capabilities for satellites.

The SE-2 utilises NVIDIA modules to maximise processing power, whilst maintaining manageable power draw. It is based on the flight proven SE-1 computer, and Spiral Blue's experience in flying 9 computers in space to date. Customisations and add-ons are available.

PROCESSING

Chip Architecture	NVIDIA Orin NX 16GB
Processing Power	Up to 157 TOPs (INT8)
GPU	Ampere GPU 1024 CUDA cores 32 Tensor cores 2x NVDLA
CPU	8-core ARM Cortex-A78AE
RAM	16GB 3200MHz

DESIGN PARAMETERS

Size	0.25U (25x96x90mm)
Weight	~100 g (excluding heat sink and shielding)
Power Usage	Max power 25W Can be throttled to 10W or 15W
Radiation	Jetson series shown 25-50 krad TID resilience Composite radiation shielding available

CONNECTIONS

Ethernet	1x Gigabit ethernet
UART	3x UART
CAN	1x CAN
USB	2x USB 3.1 1x USB 2.0 1x USB 2.0 device mode
Serial	RS485 transceivers can be installed on UART lines as required.
GPIO	Direct access to GPIOs from Jetson module
Debug	1x debug via UART

STORAGE

Storage Options	250GB, 500GB, 1TB, 2TB options available via NVMe

AVAILABILITY, COST & EXTRAS

Availability	12-16 weeks
Flight Model Cost	$50,000 USD
Engineering Model Cost	$25,000 standalone $5,000 USD with flight model
Included Extras	40 hrs engineering support per flight model

SOFTWARE

Operating System	NVIDIA JetPack version 6.2 *Ubuntu Linux based*
Utilities	Spiral Blue utilities for control and monitoring of carrier board features.

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